

Step-By Step Guide To Select POWER MOSFETs

Metal oxide semiconducting field effect transistor, commonly known as MOSFET, is a voltage-controlled device that is normally used as a switch or amplifier. It is considered one of the most crucial electronic devices in bringing about the modern electronics revolution.

This semiconducting switch finds application in a wide variety of operations, especially where power conversion takes place. However, a power MOSFET remains one of the most common causes of failure of high-power circuits like converters and inverters. Failure of a power MOSFET can be due to various reasons, but the most frequent is selection of a wrong power MOSFET for the application. This article covers the different types of MOSFETs and factors to consider while selecting a power MOSFET

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MOSFET is a voltage-controlled 3-terminal device with source, drain, and gate as the three terminals. The basic idea of the 3-terminal device is to control the flow of current through one of its terminals by accurately controlling the voltage between the other two terminals.

MOSFETs can be categorised based on the type of doping, whether N-channel or P-channel. Both types can further be classified on the basis of operation, that is, whether it is an enhancement type MOSFET (E-MOSFET) or depletion type MOSFET. Thus, there are following four types of MOSFETs:



Fig. 1: The image represents different types of MOSFET packages